

ANIMAL CLASSIFICATION USING K-NEAREST NEIGHBOURS (KNN)

1. OBJECTIVE

The objective of this assignment is to implement and evaluate the K-Nearest Neighbours (KNN) algorithm to classify animals into different types based on biological characteristics using the Zoo dataset.

2. EXPLORATORY DATA ANALYSIS (EDA)

The dataset contains 101 animals with 17 features. Count plots were used to analyze class distribution. Correlation heatmaps identified relationships such as strong correlation between 'milk' and 'hair'.

3. OUTLIER ANALYSIS

Outliers were checked using boxplots and the IQR method on the numerical feature 'legs'. Although some values like 5 or 8 legs were outside the typical range, they represent biological variations. Therefore, no outliers were removed.

4. DATA PREPROCESSING

The 'animal name' column was removed as it does not contribute to classification. No missing values were found. StandardScaler was applied since KNN is distance-based and requires feature scaling.

5. MODEL IMPLEMENTATION

The dataset was split into 80% training and 20% testing sets. KNeighborsClassifier from scikit-learn was implemented using Euclidean distance. The optimal value $K = 5$ was selected using the Elbow Method.

6. PERFORMANCE EVALUATION

The model achieved approximately 95% accuracy on the test dataset. Precision, Recall, and F1-score values were high across most classes. The confusion matrix showed minimal misclassification.

7. DECISION BOUNDARY VISUALIZATION

Principal Component Analysis (PCA) was used to reduce dimensions to two components for visualization. The decision boundary plot demonstrates how KNN groups similar animals based on nearest neighbors.

8. INTERVIEW QUESTIONS

Q1. What are the key hyperparameters in KNN?

The key hyperparameters include: 1) K (number of neighbors), which controls bias-variance tradeoff. 2) Distance metric (Euclidean, Manhattan, Minkowski). 3) Weights (uniform or distance-based). 4) Algorithm type (auto, ball_tree, kd_tree, brute) for efficiency.

Q2. What distance metrics can be used in KNN?

Common distance metrics include Euclidean distance (most common), Manhattan distance (L1), Minkowski distance (generalized form), Hamming distance (for categorical/binary data), and Cosine distance (used in high-dimensional sparse datasets).

9. CONCLUSION

The KNN algorithm performed effectively on the Zoo dataset due to clear feature patterns and proper preprocessing. Appropriate selection of K and scaling ensured high accuracy and strong classification performance.